

RAK19011 WisBlock Dual IO Base Board with Power Slot Datasheet

Overview

Description

RAK19011 is a **WisBlock Dual IO Base Board with Power Slot** that connects **WisBlock Core**, **WisBlock Power**, and **WisBlock Modules**. It has one slot for the WisBlock Core module, one for the WisBlock Power module, two IO slots, and six sensor slots (A to F) for WisBlock Modules. There are also two **2.54 mm pitch headers** exposing all key input-output pins of the WisBlock Core, including UART, I2C, SPI, and many IO pins.

WisBlock Modules are connected to the RAK19011 WisBlock Dual IO Base Board with Power Slot via **high-speed board-to-board connectors**. They provide secure and reliable interconnection to ensure the signal integrity of each data bus. A set of screws is used for fixing the modules, which makes it reliable even in a vibrating environment. Additionally, it has a user-defined button.



TIP

If you cannot find a module that fits your IoT requirements, use the standard connectors of WisBlock to develop your specific function module. WisBlock supports open-source hardware architecture and you can find tutorials showing how to create your own [Awesome WisBlock](#)  module.

Features

- Flexible building block design enabling modular functionality and expansion.
- High-speed interconnection connectors in the WisBlock Base board ensure signal integrity.
- **Multiple headers and modules slots** for WisBlock modules
 - One (1) Core slot
 - One (1) Power slot
 - Two (2) IO slots
 - Six (6) sensor (A to F) slots
 - All key input-output pins of WisBlock Core are exposed via headers
 - Access to various communication bus via headers: I2C, SPI, UART, and USB
 - One user-defined push button switch
- **Size**
 - 60 mm x 67 mm

Specifications

Overview

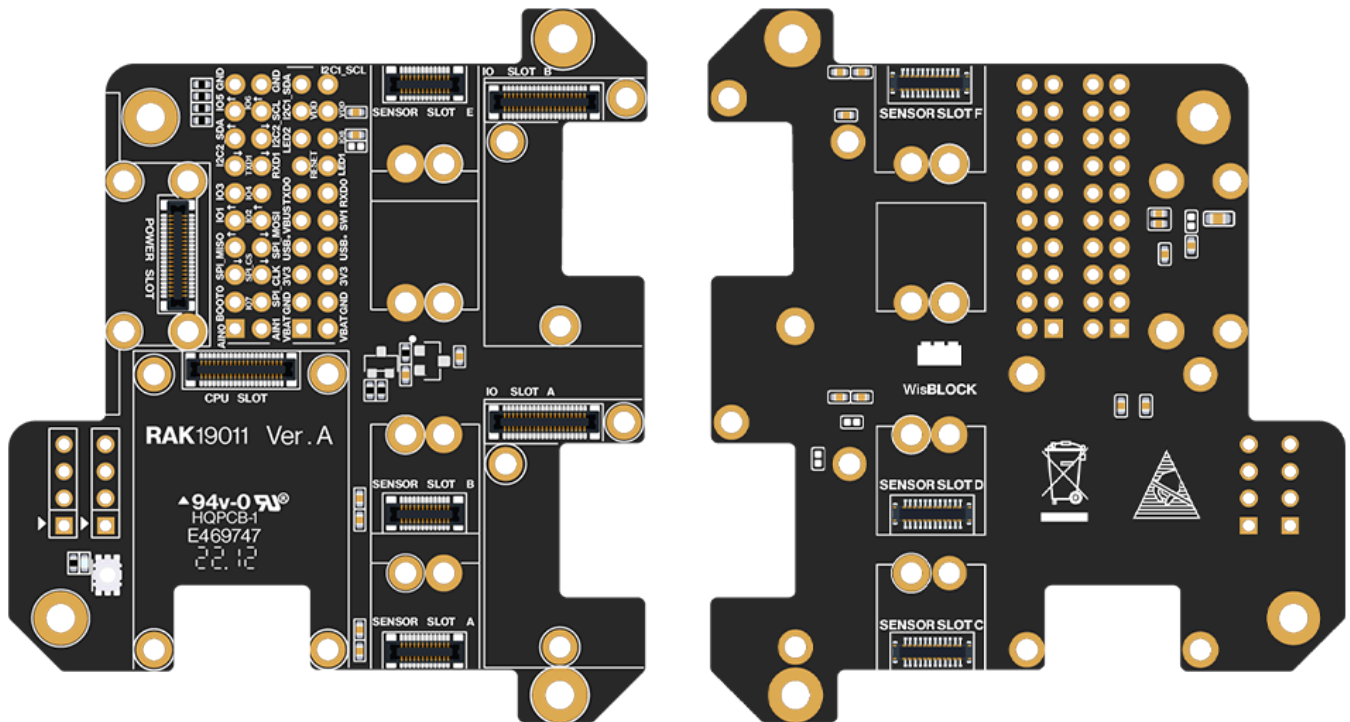


Figure 1: RAK19011 WisBlock Base top (left) and bottom (right) view

Block Diagram

The block diagram in **Figure 2** shows the internal architecture and external interfaces of the RAK19011 board.

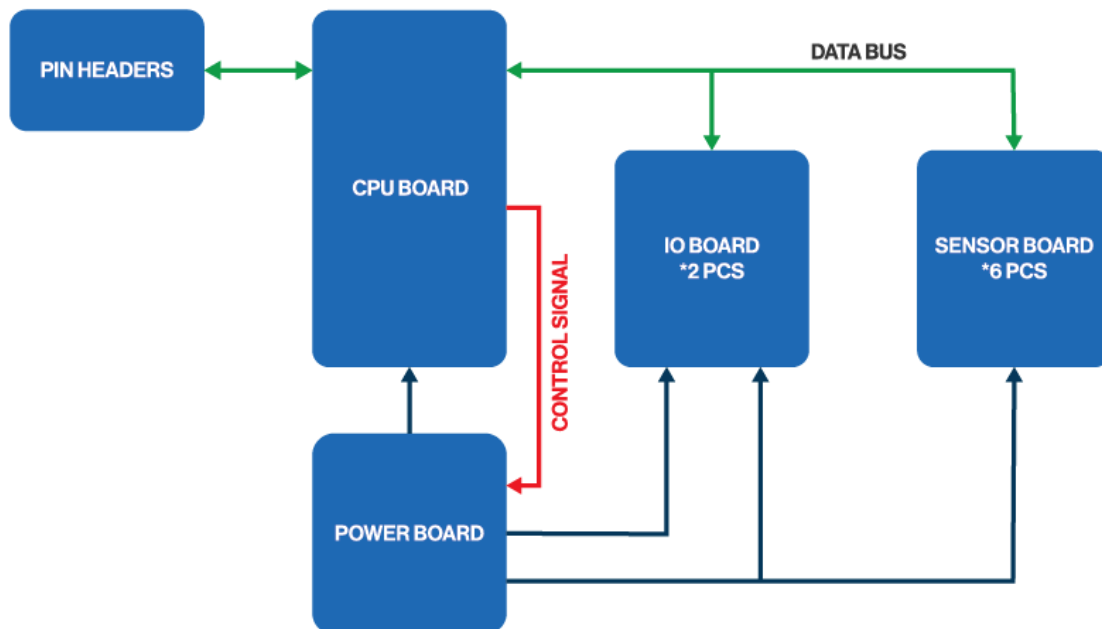


Figure 2: RAK19011 WisBlock Base block diagram

Hardware

The hardware specification is categorized into six parts. It discusses the interfacing, pinouts, and their corresponding functions and diagrams. It also covers the electrical, mechanical, and environmental parameters, including tabular data of the functionalities and standard values of the RAK19011 WisBlock Dual IO Base Board with Power Slot.

Interfaces

The RAK19011 WisBlock Dual IO Base Board with Power Slot provides the following interfaces, headers, a button, and WisConnectors:

- 1 WisBlock Core module
- 1 WisBlock Power module
- 2 IO slots for WisBlock modules
- 6 Sensor slots A-F for WisBlock modules
- 2 Headers for complete access to BOOT, I2C, SPI, UART, USB, and IO pins
- 1 User-defined button

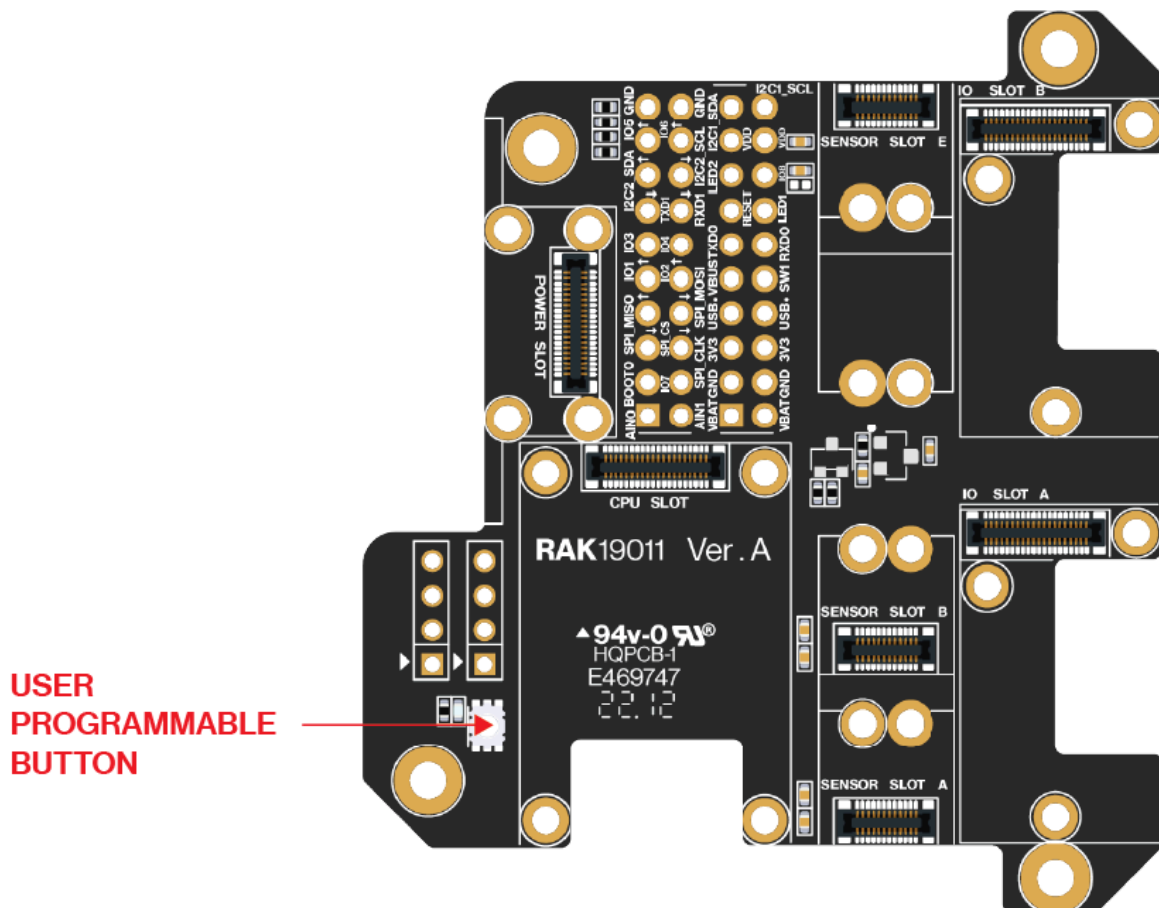


Figure 3: RAK19011 part label

J19 and J20 Headers (Core IO Pins)

On the WisBlock Dual IO Base board with power slot, there are a total of forty 2.54 mm pitch headers for IO access and extension. These IO pins are distributed to J10 and J15 pin headers, with corresponding labels on the back of the board. The pin

arrangement is based on the [40-pin WisConnector of the WisBlock Core](#).

NOTE

The BOOT pin is used for startup configuration or the sequence of the WisBlock Core connected to it. It is commonly used for uploading the bootloader and/or application firmware. The BOOT pin's state requirements depend on the specific WisBlock Core model used.

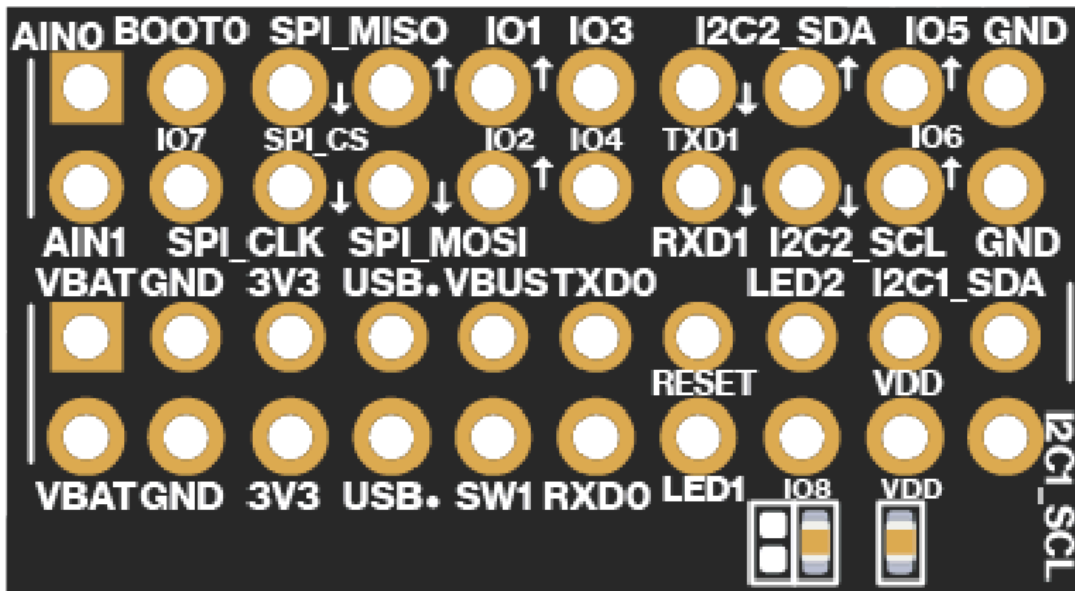


Figure 4: J19 and J20 pin header label in top side

J21 and J18 Headers (I2C and UART)

A dedicated header is available as well to have access to commonly used serial interfaces **I2C** and **UART**.

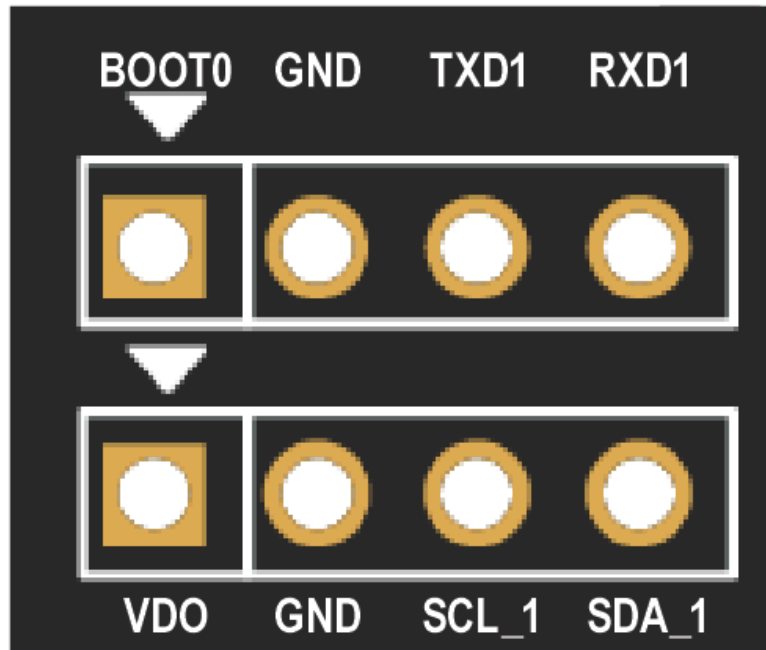


Figure 5: J21 and J18 (I2C and UART) pin header label in bottom side

User Programmable Push Button

The User Programmed Push Button shown in [Figure 3](#) is connected to the SW1 pin of the WisBlock Core.

Pin Definition

Connector for WisBlock Core

The **WisCore module connector** is a 40-pin board-to-board connector. It is a high-speed and high-density connector, with an easy attaching mechanism.

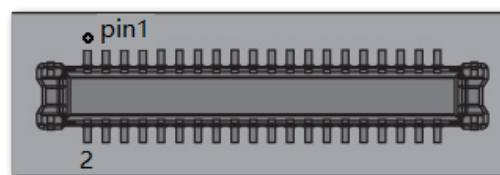


Figure 6: WisBlock Core module 40-pin connector

The table below shows the pinout of the 40-pin MCU module connector:

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
1	VBAT	2	VBAT
3	GND	4	GND

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
5	3V3	6	3V3
7	USB+	8	USB-
9	VBUS	10	SW1
11	TXD0	12	RXD0
13	RESET	14	LED1
15	LED2	16	IO8
17	VDD	18	VDD
19	I2C1_SDA	20	I2C1_SCL
21	AIN0	22	AIN1
23	BOOT0	24	IO7
25	SPI_CS	26	SPI_CLK
27	SPI_MISO	28	SPI_MOSI
29	IO1	30	IO2
31	IO3	32	IO4
33	TXD1	34	RXD1
35	I2C2_SDA	36	I2C2_SCL
37	IO5	38	IO6
39	GND	40	GND

As for the following table, it shows the definition of each pin on the WisBlock Core connector:

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery

Pin Number	Pin Name	Type	Description
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3	S	3.3 V power supply
7	USB+	I/O	USB D+
8	USB-	I/O	USB D-
9	VBUS	S	USB VBUS
10	SW1	I/O	User Defined Button (available on RAK4631/RAK4631-R and 11200 WisBlock Cores)
11	TXD0	I/O	MCU UART0 TX signal
12	RXD0	I/O	MCU UART0 RX signal
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charging indication
15	LED2	I/O	LED for custom usage
16	LED3	I/O	LED for custom usage
17	VDD	S	Generated by MCU module for power sensor board if the MCU IO level is not 3.3 V
18	VDD	S	Generated by MCU module for power sensor board if the MCU IO level is not 3.3 V
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signals
21	ADC_VBAT	A	Analog input for ADC (Connected to a battery)
22	AIN1	A	Analog input for ADC

Pin Number	Pin Name	Type	Description
23	BOOT0	I	For ST MCU only. The MCU will enter boot mode if this pin is connected to VDD.
24	IO7	I/O	Not connected
25	SPI_CS	I/O	SPI chip select signal
26	SPI_CLK	I/O	SPI clock signal
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	IO1	I/O	General purpose IO
30	IO2	I/O	Used for 3V3_S enable
31	IO3	I/O	General purpose IO
32	IO4	I/O	General purpose IO
33	TXD1	I/O	MCU UART1 RX signal
34	RXD1	I/O	MCU UART1 RX signal
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Connector for WisBlock Power Module

The **WisPower module connector** is a 40-pin board-to-board connector. It is a high-speed and high-density connector, with an easy attaching mechanism.

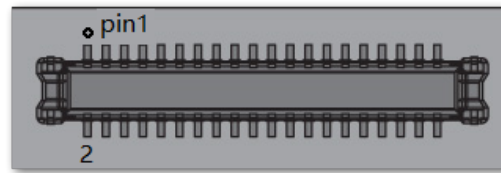


Figure 7: WisBlock Core module 40-pin connector

The table below shows the pinout of the 40-pin Power module connector:

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
1	VBAT	2	VBAT
3	GND	4	GND
5	3V3	6	3V3
7	USB+	8	USB-
9	VBUS	10	VBUS
11	NC	12	NC
13	RESET	14	LED1
15	LED2	16	NC
17	3V3	18	3V3
19	I2C1_SDA	20	I2C1_SCL
21	AIN0	22	AIN1
23	NC	24	NC
25	SPI_CS	26	SPI_CLK
27	SPI_MISO	28	SPI_MOSI
29	NC	30	NC
31	NC	32	NC
33	NC	34	NC

Pin Number	Function Name of WisBlock Base	Pin Number	Function Name of WisBlock Base
35	I2C2_SDA	36	I2C2_SCL
37	IO5	38	IO6
39	GND	40	GND

The following table defines each pin of the WisBlock Power slot connector:

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3	S	3.3 V power supply
7	USB+	I/O	USB D+
8	USB-	I/O	USB D-
9	VBUS	S	USB VBUS
10	VBUS	I/O	USB VBUS
11	NC	NC	Not connected
12	NC	NC	Not connected
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charging indication
15	LED2	I/O	LED for custom usage
16	NC	NC	Not connected

Pin Number	Pin Name	Type	Description
17	3V3	S	3.3 V power supply
18	3V3	S	3.3 V power supply
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signal
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	NC	NC	Not connected
24	NC	NC	Not connected
25	SPI_CS	I/O	SPI chip select signal
26	SPI_CLK	I/O	SPI clock signal
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	NC	NC	Not connected
30	NC	NC	Not connected
31	NC	NC	Not connected
32	NC	NC	Not connected
33	NC	NC	Not connected
34	NC	NC	Not connected
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO

Pin Number	Pin Name	Type	Description
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Connectors for WisBlock Sensor

The WisBlock sensor module connector is a 24-pin board-to-board connector.

⚠ WARNING

The WisBlock 24-pin connectors have the same connections for **3V3_S**, **GND**, **I2C**, and **SPI**. However, **UART** and **IO** pins are not the same for all slots.

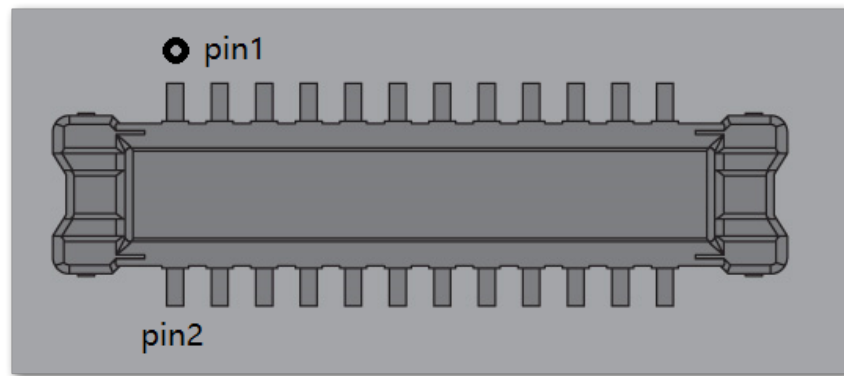


Figure 8: WisBlock 24-pin module connector

Pinout definition for standard size slot (A-D):

D	C	B	A	Pin Number	Pin Number	A	B	C	D
NC	NC	NC	TXD0	1	2	GND	GND	GND	GND
SPI_CS	SPI_CS	SPI_CS	SPI_CS	3	4	SPI_CLK	SPI_CLK	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	SPI_MISO	SPI_MISO	5	6	SPI_MOSI	SPI_MOSI	SPI_MOSI	SPI_MOSI
I2C1_SCL	I2C1_SCL	I2C1_SCL	I2C1_SCL	7	8	I2C1_SDA	I2C1_SDA	I2C1_SDA	I2C1_SDA
VDD	VDD	VDD	VDD	9	10	IO2	IO1	IO4	IO6
3V3_S	3V3_S	3V3_S	3V3_S	11	12	IO1	IO2	IO3	IO5

D	C	B	A	Pin Number	Pin Number	A	B	C	D
NC	NC	NC	NC	13	14	3V3_S	3V3_S	3V3_S	3V3_S
NC	NC	NC	NC	15	16	VDD	VDD	VDD	VDD
NC	NC	NC	NC	17	18	NC	NC	NC	NC
NC	NC	NC	NC	19	20	NC	NC	NC	NC
NC	NC	NC	NC	21	22	NC	NC	NC	NC
GND	GND	GND	GND	23	24	RXD0	NC	NC	NC

Pinout definition for double-size sensor slot (E and F):

Connector F	Connector E	Pin Number	Pin Number	Connector E	Connector F
TXD1	TXD0	1	2	GND	GND
SPI_CS	SPI_CS	3	4	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	5	6	SPI_MOSI	SPI_MOSI
I2C1_SCL	I2C1_SCL	7	8	I2C1_SDA	I2C1_SDA
VDD	VDD	9	10	IO3	IO5
3V3_S	3V3_S	11	12	IO4	IO6
NC	NC	13	14	3V3_S	3V3_S
NC	NC	15	16	VDD	VDD
NC	NC	17	18	NC	NC
NC	NC	19	20	NC	NC
NC	NC	21	22	NC	NC
GND	GND	23	24	RXD0	RXD1

The following table shows the pin name and description for each pin on the 24-pin WisBlock module connector.

Pin Number	Pin Name	Type	Description
1	TXD1	I/O	UART TX signal
2	GND	S	Ground
3	SPL_CS	I/O	SPI chip select signal
4	SPL_CLK	I/O	SPI clock signal
5	SPL_MISO	I/O	SPI MISO signal
6	SPL_MOSI	I/O	SPI MOSI signal
7	I2C1_SCL	I/O	I2C clock signal
8	I2C1_SDA	I/O	I2C data signal
9	VDD	S	Generated by CPU module. Used to power sensor board if MCU IO level is not 3.3 V
10	IOx	I/O	General purpose IO pin. When 3V3_S is used, this pin cannot be used as an interrupt input.
11	3V3_S	S	3.3 V power supply. This power pin is controlled by IO2 from the WisBlock Core module.
12	IOx	I/O	General purpose IO pin. When 3V3_S is used, this pin cannot be used as an interrupt input.
13	NC	NC	Not connected
14	3V3_S	S	3.3 V power supply. This power pin is controlled by IO2 from the WisBlock Core module.
15	NC	NC	Not connected
16	VDD	S	Generated by CPU module. Used to power sensor board if the MCU IO level is not 3.3 V.
17	NC	NC	Not connected

Pin Number	Pin Name	Type	Description
18	NC	NC	Not connected
19	NC	NC	Not connected
20	NC	NC	Not connected
21	NC	NC	Not connected
22	NC	NC	Not connected
23	GND	S	Ground
24	RXD1	I/O	UART RX signal

Connector for WisBlock IO Slot

The WisBlock Module IO Slot connector, as shown in **Figure 9**, is a 40-pin board-to-board connector.

NOTE

The two WisBlock 40-pin connectors have the same connections for all I/O, signal, and serial pins (UART, SPI, I2C).

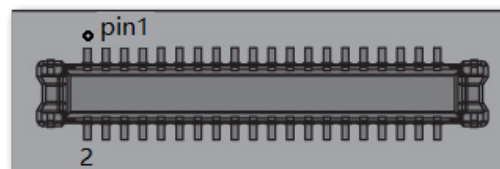


Figure 9: WisBlock IO slot connector

Pinout definition for IO slot:

Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B
VBAT	VBAT	1	2	VBAT	VBAT
GND	GND	3	4	GND	GND
3V3	3V3	5	6	3V3_S	3V3_S
USB+	USB+	7	8	USB-	USB-
VBUS	VBUS	9	10	SW1	SW1

Connector B	Connector A	Pin Number	Pin Number	Connector A	Connector B
TXD0	TXD0	11	12	RXD0	RXD0
RESET	RESET	13	14	LED1	LED1
LED2	LED2	15	16	LED3	LED3
VDD	VDD	17	18	VDD	VDD
I2C1_SDA	I2C1_SDA	19	20	I2C1_SCL	I2C1_SCL
AIN0	AIN0	21	22	AIN1	AIN1
NC	NC	23	24	NC	NC
SPI_CS	SPI_CS	25	26	SPI_CLK	SPI_CLK
SPI_MISO	SPI_MISO	27	28	SPI_MOSI	SPI_MOSI
IO1	IO1	29	30	IO2	IO2
IO3	IO3	31	32	IO4	IO4
TXD1	TXD1	33	34	RXD1	RXD1
I2C2_SDA	I2C2_SDA	35	36	I2C2_SCL	I2C2_SCL
IO5	IO5	37	38	IO6	IO6
GND	GND	39	40	GND	GND

The following table shows the pin name and description of the WisBlock IO module connector.

Pin Number	Pin Name	Type	Description
1	VBAT	S	Power supply from battery
2	VBAT	S	Power supply from battery
3	GND	S	Ground

Pin Number	Pin Name	Type	Description
4	GND	S	Ground
5	3V3	S	3.3 V power supply
6	3V3_S	S	3.3 V power supply. Can be shut down by a CPU module.
7	USB+	I/O	USB D+
8	USB-	I/O	USB D-
9	VBUS	S	5 V input for USB
10	SW1	I/O	User Defined Button (available on RAK4631/RAK4631-R and 11200 WisBlock Cores)
11	TXD0	I/O	MCU UART0 TX signal
12	RXD0	I/O	MCU UART0 RX signal
13	RESET	I	Connected to the reset switch, for MCU reset
14	LED1	I/O	LED for battery charge indicator
15	LED2	I/O	LED for custom used
16	LED3	I/O	LED for custom used
17	VDD	S	Generated by CPU module - Used for power sensor board if the MCU IO level is not 3.3 V
18	VDD	S	Generated by CPU module - Used for power sensor board if the MCU IO level is not 3.3 V.
19	I2C1_SDA	I/O	The first set of I2C data signal
20	I2C1_SCL	I/O	The first set of I2C clock signal
21	AIN0	A	Analog input for ADC
22	AIN1	A	Analog input for ADC
23	NC	NC	Not connect

Pin Number	Pin Name	Type	Description
24	NC	NC	Not connect
25	SPI_CS	I/O	SPI chip select signal
26	SP_CLK	I/O	SPI clock
27	SPI_MISO	I/O	SPI MISO signal
28	SPI_MOSI	I/O	SPI MOSI signal
29	IO1	I/O	General purpose IO
30	IO2	I/O	Used for 3V3_S enable
31	IO3	I/O	General purpose IO
32	IO4	I/O	General purpose IO
33	TXD1	I/O	MCU UART1 TX signal
34	RXD1	I/O	MCU UART1 RX signal
35	I2C2_SDA	I/O	The second set of I2C data signal
36	I2C2_SCL	I/O	The second set of I2C clock signal
37	IO5	I/O	General purpose IO
38	IO6	I/O	General purpose IO
39	GND	S	Ground
40	GND	S	Ground

Electrical Characteristics

Absolute Maximum Ratings

The absolute maximum ratings of the device are shown in the table below. Stress ratings are for the functional operation of the device.



1. If the stress rating exceeds the listed value, it may cause permanent damage to the device.
2. Exposure to maximum-rated conditions may affect device reliability.

Ratings	Maximum Value	Unit
IOs of WisConnector	-0.3 to VDD+0.3	V
ESD	2000	V

NOTE

The RAK19011, like any electronic equipment, is sensitive to **electrostatic discharge (ESD)**. Improper handling can cause permanent damage to the module.

Current Consumption

The RAK19011 is designed for **low-power IoT products**. Its power supply uses a high-efficiency low ground current regulator. When there is no module on RAK19011, the **leakage current is lower than 2 μ A**. With WisBlock Core and WisBlock sensor on it, the sleep current is **lower than 10 μ A**. When a LoRa module is transmitting, the current may reach **130 mA**.

Conditions	Current	Unit
Leakage current, without any module on RAK19011	2	μ A
Idle current, with WisBlock Core and WisBlock Modules in sleep mode	10	μ A
Working current, with LoRa module transmitting	130	mA

Mechanical Characteristics

Board Dimensions

NOTE

- You may also refer and download the [M1.2 Stand-off fastener/inserts datasheet](#) .

Figure 10 shows the detailed mechanical dimensions of RAK19011.

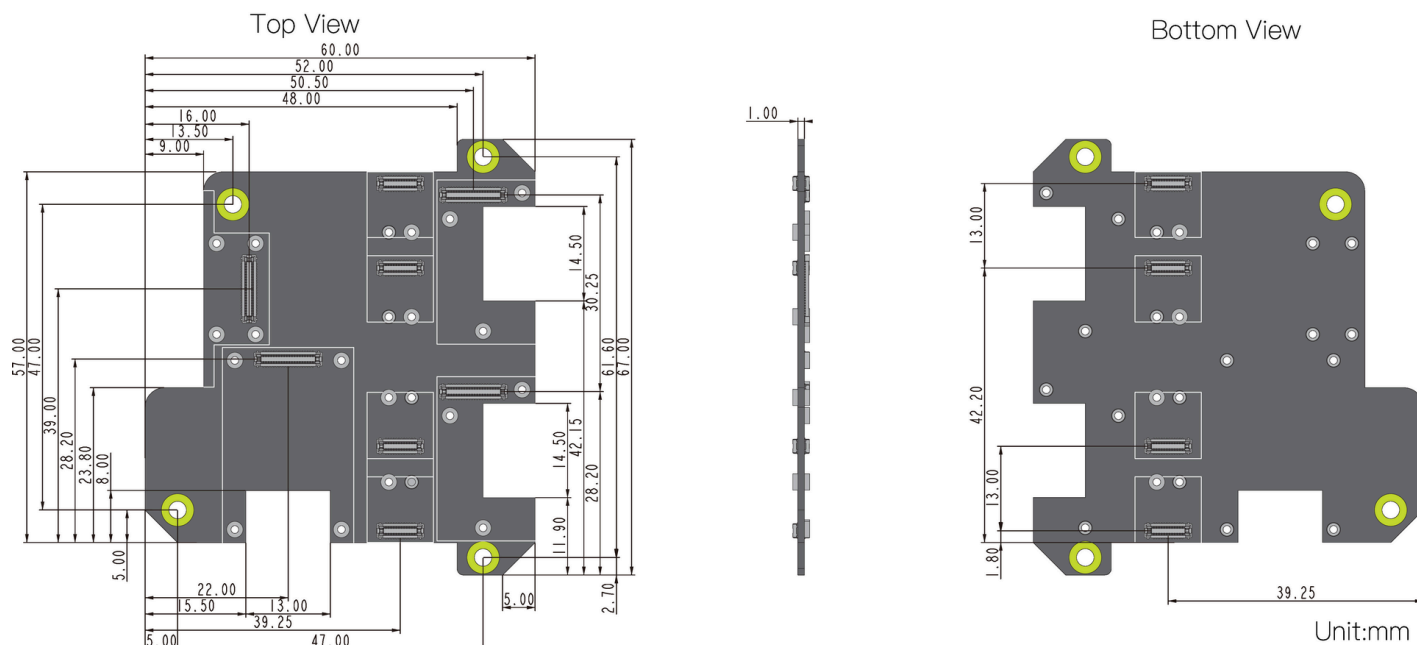


Figure 10: RAK19011 board dimensions

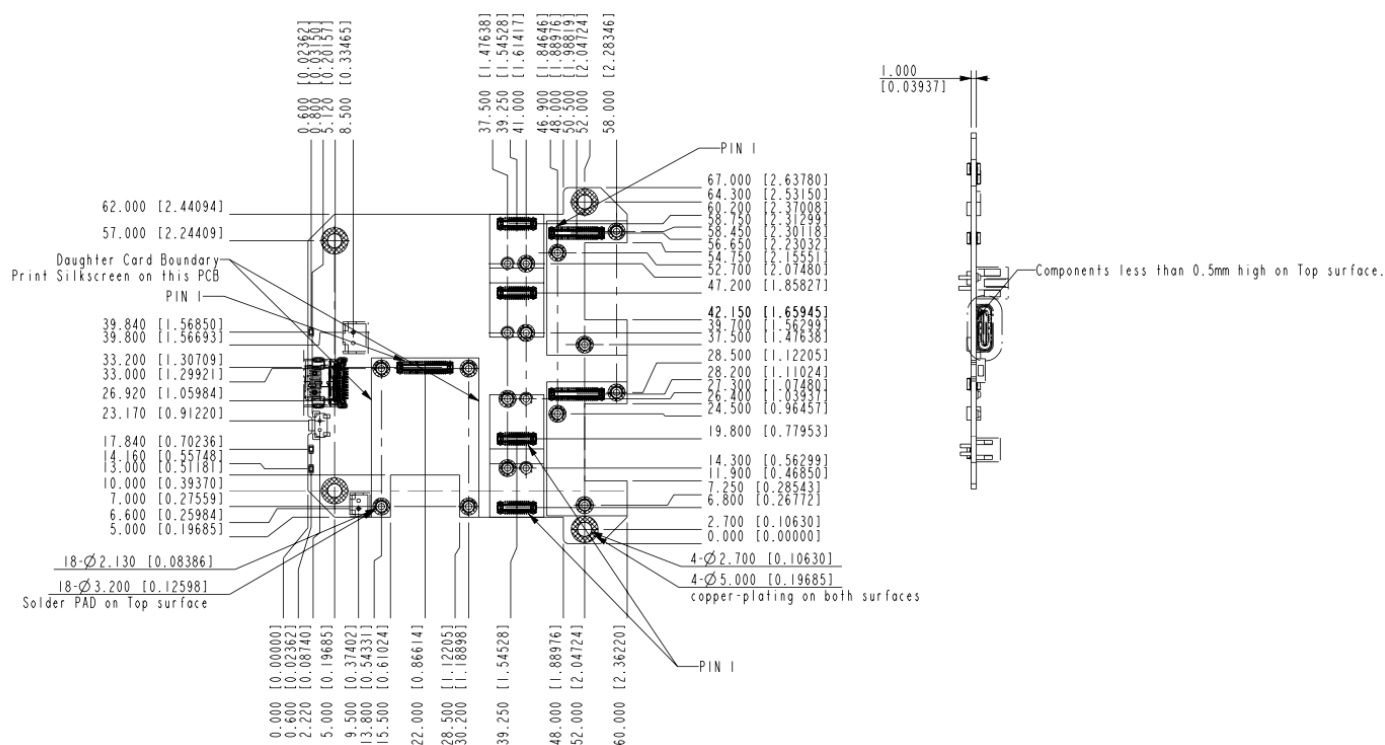


Figure 11: RAK19011 mechanical dimensions

WisConnector PCB Layout

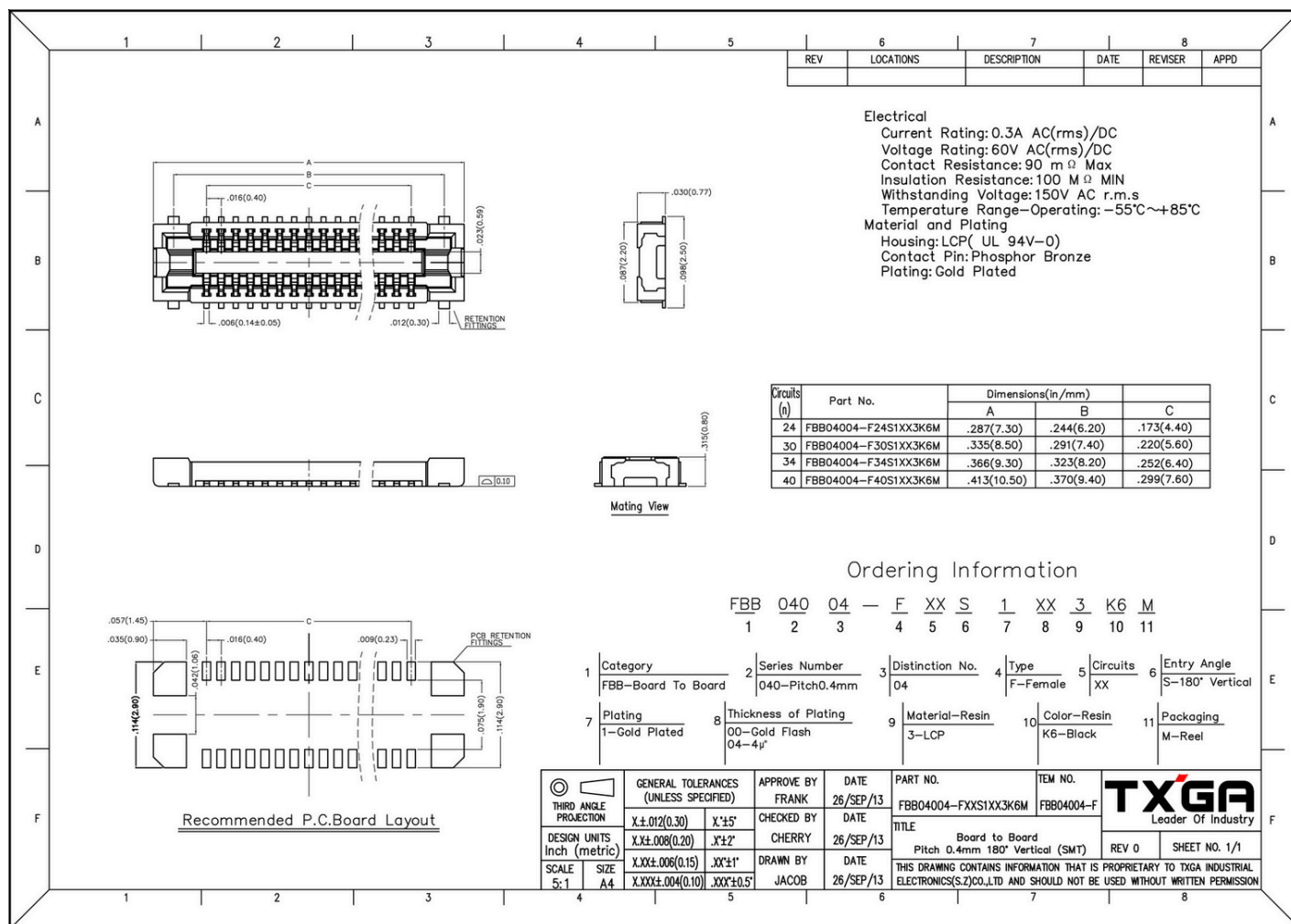


Figure 12: WisConnector PCB footprint and recommendations

Environmental Characteristics

The table below lists the operation and storage temperature requirements of RAK19011:

Parameter	Minimum	Typical	Maximum
Operational temperature range	-35 °C	+25 °C	+75 °C
Extended temperature range	-40 °C	+25 °C	+80 °C
Storage temperature range	-40 °C	+25 °C	+80 °C

Schematic Diagram

The component schematic diagram of the RAK19011 is shown in Figure 13 and Figure 14.

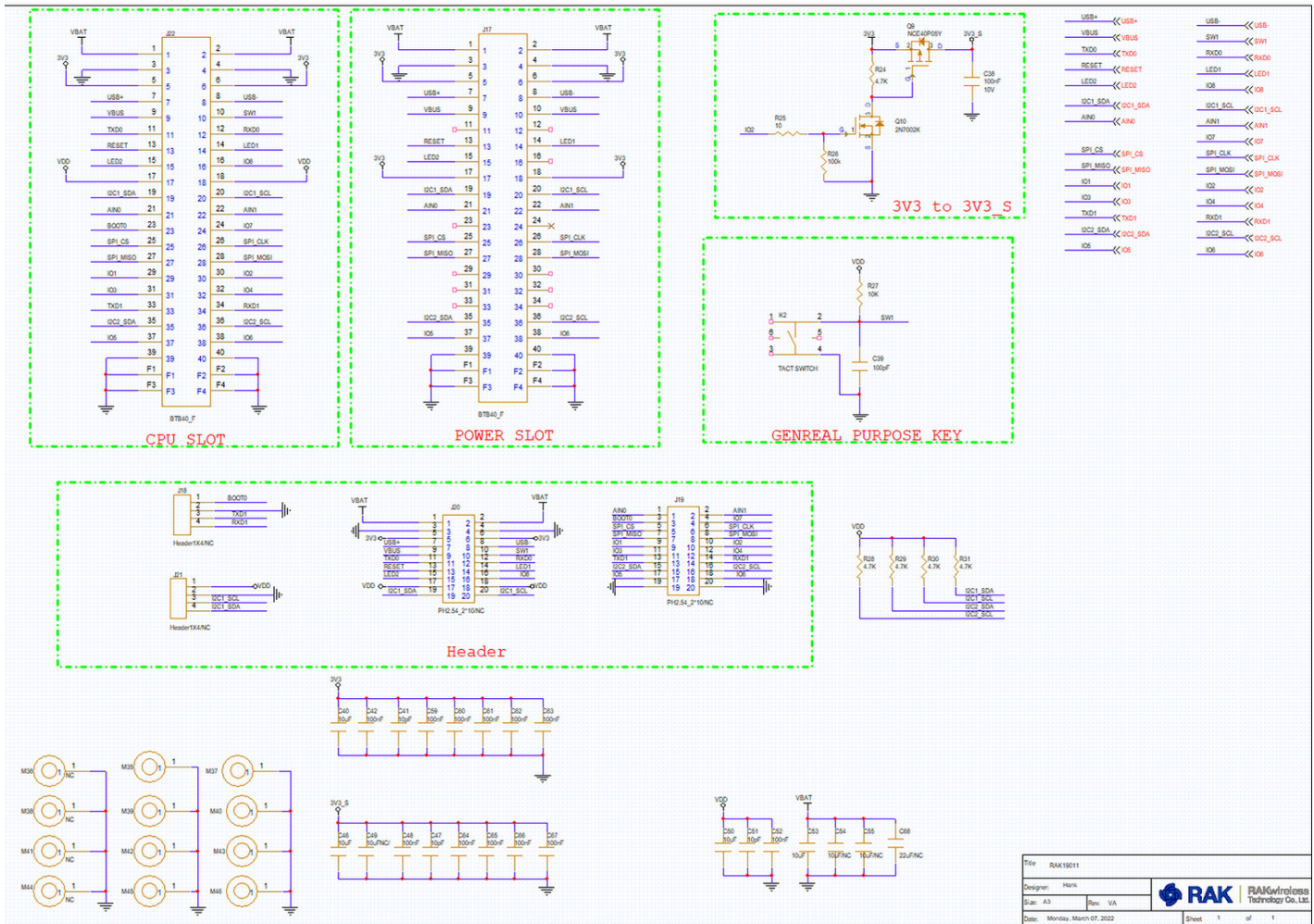


Figure 13: RAK19011 schematic diagram 1

